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Fully and Partially Coherent Scatterers Detection: Theoretical Analysis and Performance Comparison on Very High Resolution SAR Data

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***Abstract*—Persistent Scatterers (PS) and Distributed Scatterers (DS) detection is a key step in Multitemporal Differential Interferometric Synthetic Aperture Radar (MT-DInSAR) processing for the monitoring from the space of slow displacements of ground scatterers on the Earth. Available detection approaches - Generalized Likelihood Ratio Test (GLRT), Squeeze-SAR (SqueeSAR) and Component Analysis and Selection SAR (CAE-SAR) based detectors - were all designed assuming the absence of decorrelation, i.e. they implement a Fully-Coherent Detection (FCD) scheme. Starting from the theoretical analysis available in the literature, in this work we derive a Partially-Coherent Detection (PCD) scheme able to cope with single and (unfiltered and filtered) multilook data. We compare, under a fixed false alarm level, FCD and PCD schemes on both simulated and real data associated to a X-Band COSMO-SkyMed interferometric dataset. The work aims to provides for the first time evidences on experimental data of the effect of weighting of complex interferometric pairs in the process of PS and DS detection. Presented results show that, accounting for scattering coherence according to well-known, albeit basic angular and temporal decorrelation models, may lead to improvements in the detection of PS and DS, at both full and reduced resolution.**

***Index Terms*—Persistent Scatterer, Distributed Scatterers, Persistent Scatterers Interferometry, SAR Interferometry, SAR Tomography, Detection**

I. Introduction

Persistent Scatterers Interferometry (PSI) is a well-developed Multitemporal Differential Interferometric Synthetic Aperture Radar (MT-DInSAR) technique used to monitor surface displacements [1]–[5]. It leverages on the capability of some natural scatterers to maintain a coherent response over the time (scatterer persistence).

Based on the characteristics of the imaged scene, such natural scatterers, referred to as PS or DS depending on whether they are concentrated or distributed, generally constitute only a low percentage of the total number of imaged pixels [6]. Existing techniques must therefore distinguish pixels entailing PS/DS from noisy ones [1], [7]. When this separation is carried out at full resolution (single-look - SL) or near full-resolution (multilook - ML) for PS or DS, respectively, it requires identifying a statistic capable of measuring the degree of coherence of the scatterers response along the time (across the images composing the interferometric data stack) i.e., their "persistence".

The first and most widely used technique, PS-InSAR™, identifies PS by comparing the measured phase vector of the multi-temporal image stack with a phase model that depends on scatterer parameters, hereafter referred to as the Parameters of Interest (PoI) [2]. This operation, performed in the complex domain, measures the so-called Multitemporal Interferometric Coherence (MIC) [1], [2], [8]. As with all approaches considered in this work, it requires a phase calibration step, at least to compensate the atmospheric disturbance. Without loss of generality, we consider interferometric data processing procedures involving two steps [9], [10]. The first step, implemented at a reduced resolution, estimates and compensate the atmospheric phase delay and possible small-scale (low resolution) displacements. The second step performs the PS/DS identification, along with the estimation of the associated PoI, typically restricted (as in this work) to Residual Topography (RT) and (residual) Displacement Mean Velocity (DMV) (displacement rate). Well-known aspects of the first stage of the MT-DInSAR processing - including image registration, external Digital Elevation Model (DEM) phase compensation and phase unwrapping - are not addressed here and are instead referred to the extensive existing literature [11].

PS identification, as applied to single-look data, has been subsequently framed in the detection theory context. Tomographic-based detectors have been developed; in particular, the Single Look Generalized Likelihood Ratio Test (SL-GLRT) in [8], derived under the assumption that the PS response is superimposed to an additive white Gaussian noise, has been shown to improve PS detection compared to the aforementioned (phase-only) model-matching (MIC) under a fixed Probability of False Alarm (PFA). Tomographic approaches have also enabled the extension of PS order. Indeed, several relevant contributions in the literature have addressed the separation of multiple scatterers interfering in the same pixel. This topic, relevant to the so-called layover solution which is important for urban applications, [12]–[17], have also benefited of the application of Compressed Sensing techniques for the efficient estimation of the PoI of interfering scatterers [18]–[20]. Nevertheless, this aspect is beyond the scope of the present work. Within the framework of the Small Baselines Subsets (SBAS) approach, PS are identified at single-look in [9] by exploiting only a subset of all possible full-resolution interferograms selected by limiting with an empirical criterion

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the temporal and spatial baselines. Recent developments based on artificial neural networks detectors (e.g, [21], [22]), which relax the fixed PFA assumption, are also beyond the scope of this work.

Decorrelation is a major limiting factor that can significantly impair the PS detection at single-look (full-resolution), especially in non-urban areas where scattering is typically weak. Frequent changes of surface scattering properties over the time and/or the spatial distribution of the scattering within the resolution cell, lead to temporal and geometrical decorrelation, respectively [23]. The term DS is used to describe such condition, indicating scatterers distributed within a resolution cell, but also distributed (with the same statistical characterization) over adjacent pixels. Strictly speaking the term "distributed" refers to geometrical decorrelation rather than temporal decorrelation. To address this, DS detection schemes leveraging multilook processing have been proposed. Multilooking improves the interferometric phase quality at the expense of a spatial resolution loss, thereby increasing the density of detectable pixels. For example, the SqueeSAR [6] approach extends DS identification by estimating, from the multilooked sample covariance matrix, the phase vector corresponding to an equivalent PS. The CAESAR method [24], based on the adoption of the Principal Component Analysis (PCA), extends this concept to the full complex domain, enabling also the identification of multiple scattering mechanisms. Both methods implement a "diadic approximation" of the sample covariance matrix, effectively filtering the data. CAESAR achieves this by extracting the dominant component from the eigen-decomposition of the sample covariance matrix, while SqueeSAR preforms a quasi maximum-likelihood estimation of the phase response associated to the equivalent PS obtained by "squeezing" the covariance matrix. Detection schemes applied to the filtered data, namely CAESAR-D (CAESAR+SL-GLRT) and SqueeSAR-D (SqueeSAR+MIC), have been analyzed in [25] and [26]. Both methods largely outperform the SL-GLRT in non urban areas, with the CAESAR-D showing a slight performance advantage over SqueeSAR-D, [26]. Interestingly, CAESAR (and similarly SqueeSAR) employs a detection threshold independent of the number of looks, simplifying implementation in in multi-resolution analysis with adapting multilooking.

Both CAESAR and SqueeSAR have been show to be upper-bounded in detection performances by the Multi Look Generalized Likelihood Ratio Test (MGLRT) [27], which exploits the whole sample covariance matrix rather than its approximations. Still, a disadvantage of MGLRT is that, in multi-resolution analysis, the detection threshold must be adjusted locally based on the number of averaged looks [25]. Moreover, phase miscalibrations can be less relevant for the detection, see [25] for a thorough comment on this characteristic.

Except for SBAS, which implements a PS detection at full resolution, the above mentioned DS detection schemes differ in the employed data component, but all assume the presence of a fully coherent scattering mechanism. They will be here referred to as Fully-Coherent Detection (FCD) schemes. Note that the term "coherent" detection refers to the presence of correlation across the multi-temporal stack of acquisitions on a given look. This differs from a similar term used in [28], which describes a possible but seldom occurring condition of spatial correlation, i.e. along the looks.

The problem of extracting information from partially coherent scatterers within the PSI framework was firstly addressed in [29]. Furthermore, a Partially-Coherent Detection (PCD) scheme that explicitly incorporates the correlation properties of the scatter at the detection stage was presented in [30], where a single-look formulation, referred to as Single-Look Partially-Coherent Detection (SL-PCD), was derived and analyzed theoretically. Finally, a preliminary work toward a PCD scheme operating with multilook was presented in [31].

The contribution of this work is a deeper analysis of the detection of decorrelating scatterers, still referred to as DS (with "D" intentionally denoting both "Decorrelating" and "Distributed" within the pixel and over adjacent pixels). To this end, first of all in Section II we unify the detection schemes proposed in the literature under a common umbrella. This includes the SL-PCD (and, as a particular case, the SBAS detector in [9]), SL-GLRT and MGLRT for PS and DS detection, respectively, as well as multilook detectors based on data filtering, i.e., CAEASAR-D and SqueeSAR-D. Secondly, following the Rao test criteria, we extend the SL-PCD to the multilook case. The resulting Multilook Partially-Coherent Detection (ML-PCD) scheme involves a proper weighting of the elements of the sample covariance matrix, which includes all interferometric measurements. Thirdly, based on the same approach used to eliminate the MGLRT's need for adaptive threshold selection in multi-resolution analyses, we also introduce a CAESAR-based PCD. This allows extending CAESAR-D (and hence SqueeSAR-D) beyond the fully coherent framework. With the aim of testing the above detection schemes on real data, decorrelation is subsequently modeled in Section III as a combination of independent sources, using a triangular model for spatial decorrelation [23] and an exponential decay model for the temporal decorrelation [32]. After the description of a strategy for the estimation of the decorrelation parameters, a performance assessment and comparison of all the PCD and FCD schemes are provided using simulated data in Section IV and a real dataset acquired by the COSMO-SkyMed constellation in Section V. Further remarks and comments on the experimental results are provided in Section VI, before the conclusions in Section VII. As a final consideration, it is important to note that the overall analysis, model simplifications, and experimental setup adopted in this work are primarily oriented to the main objective, which is not to develop a ready-to-use operational tool optimized for maximum performance, but rather to investigate and highlight, for the first time to the best of the authors' knowledge, the effects of weighting interferograms in the detection of decorrelating scatterers.

II. Signal Model and Detection Schemes

Let us consider a stack of N co-registered and phase-calibrated SAR images acquired at times $t_1, \ldots, t_N$ and with orthogonal baselines $b_1, \ldots, b_N$ with respect to a reference (master) acquisition. Furthermore, for a given pixel, let us

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suppose the presence of a scatterer characterized by a set of PoI. As stated in the introduction, without lack of generality, we consider the RT, indicated by s, and the DMV, denoted by v. Under these conditions, the complex data vector $\boldsymbol{x}$ of length N, which collects the pixel values across the images, can be modeled in the most general form as:

$$\boldsymbol{x} = \boldsymbol{a}(s,v) \odot \boldsymbol{\gamma} + \boldsymbol{n} \tag{1}$$

where $\odot$ is the Hadamard (element-wise) product operator. In (1), $\boldsymbol{\gamma}$ and $\boldsymbol{n}$ are the complex vectors collecting the scatterer's backscattering coefficients and the additive noise contribution, respectively, at the different antennas. Furthermore, $\boldsymbol{a}(s,v)$ is the so called *steering vector*, which stacks the scatterer's phase signature across the different acquisitions. Its k-th element is given by:

$$a_k(s,v) = \frac{\exp\left[-j2\pi\left(\xi_k s + \zeta_k v\right)\right]}{\sqrt{N}} \tag{2}$$

with

$$\xi_k = \frac{2b_k}{\lambda r_0 \sin\theta} \tag{3}$$

and

$$\zeta_k = \frac{2t_k}{\lambda} \tag{4}$$

where λ is the operating wavelength, while r_0 and θ are the range and look angles, respectively, relevant to the considered pixel.

The observation $\boldsymbol{x}$ is assumed to be the realization of a random vector that, with a slight abuse of notation, we still denote as $\boldsymbol{x}$. The scatterer correlation properties along the antennas are thus taken into account by the covariance matrix $\mathbf{C}_\gamma$ of the random vector $\boldsymbol{\gamma}$, usually assumed to be zero-mean circularly-symmetric Gaussian distributed. This matrix is given by:

$$\mathbf{C}_\gamma = E\left(\boldsymbol{\gamma}\boldsymbol{\gamma}^H\right) = \sigma_\gamma^2 \mathbf{M} \tag{5}$$

where $E(\cdot)$ and $(\cdot)^H$ are the statistical expectation and the hermitian (transpose and conjugate) operators, respectively. Parameter σ_γ^2 is the variance of the elements of $\boldsymbol{\gamma}$ and $\mathbf{M}$ is the normalized (all the elements along the diagonal are equal to 1) real valued symmetric matrix sometimes referred to as *structure matrix*, since it defines the coherence structure of the scatterer, [30]. The two limiting cases of fully coherent and fully incoherent scattering corresponds to a (rank-one) constant structure matrix and to a (full-rank) diagonal structure matrix, respectively. All intermediate levels of coherence result in a structure matrix that properly models the decorrelation phenomena affecting the scatterer, such as temporal decorrelation and (surface or volumetric) angular decorrelation. The overall decorrelation of the signal, which also accounts for the noise contribution introduced by the system, is described by the covariance matrix $\mathbf{C}_x$ of the entire data vector (1), given by:

$$\mathbf{C}_x = E\left(\boldsymbol{x}\boldsymbol{x}^H\right) = \sigma_\gamma^2 \mathbf{M} \odot \mathbf{A}(s,v) + \sigma_n^2 \mathbf{I}_N \tag{6}$$

where $\mathbf{A}(s,v) = \boldsymbol{a}(s,v)\boldsymbol{a}^H(s,v)$, $\mathbf{I}_N$ is the $N \times N$ identity matrix, and σ_n^2 is the variance of the components of the random vector $\boldsymbol{n}$, assumed to be distributed as a Gaussian white noise and uncorrelated with $\boldsymbol{\gamma}$.

The problem of selecting and reliably characterizing PS/DS on a pixel-by-pixel basis is approached within the framework of the detection theory, by formulating a binary hypothesis test aimed at discriminating between the absence (hypothesis $\mathcal{H}_0$) and the presence (hypothesis $\mathcal{H}_1$) of a scatterer in the tested pixel. The solution that unifies all the detection schemes previously proposed in the literature, for known PoI, is given by the following decision rule:

$$\frac{\mathrm{tr}\left\{\left[\mathbf{M} \odot \mathbf{A}(s,v)\right]\widehat{\mathbf{C}}_x\right\}}{\mathrm{tr}\left(\widehat{\mathbf{C}}_x\right)} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{7}$$

where $\mathrm{tr}(\cdot)$ denote the matrix trace operator and T is the detection threshold, set according to the desired level of false alarm. Furthermore, $\widehat{\mathbf{C}}_x$ is the data sample covariance matrix that, when appropriately specialized, can accommodate unfiltered single-look and multilook data as well as CAESAR/SqueeSAR filtered multilook data.

Since the scatterers PoI are always unknown in practice, they are estimated by maximizing the numerator of (7):

$$(\hat{s},\hat{v}) = \arg\max_{s,v} \mathrm{tr}\left\{\left[\mathbf{M} \odot \mathbf{A}(s,v)\right]\widehat{\mathbf{C}}_x\right\} \tag{8}$$

which, under the assumption of Gaussian distributed data, is the Maximum Likelihood Estimate (MLE) of the PoI when $\mathbf{M}$ has all one entries, i.e. for FCD schemes, and $\widehat{\mathbf{C}}_x$ is obtained from unfiltered data, [8], [27].

It is worth emphasizing that the statistic on the left hand side of the test (7) represents the normalized Frobenius inner product between the model component $\mathbf{M} \odot \mathbf{A}(\hat{s},\hat{v})$, accounting for the coherence structure and the estimated signature of the tested scatterer, and data component $\widehat{\mathbf{C}}_x$, derived from the observed and possibly filtered data. Note that the model component does not appear in the normalization, as its trace is equal to 1.

As for the model component, it can be designed to adapt the rule (7) to the coherence properties of the scatterer to be detected. On one hand, FCD schemes are obtained by setting $\mathbf{M}$ to the matrix composed entirely of ones. On the other hands, PCD schemes can be developed by appropriately modeling the structure matrix. Regarding the data component, in this work we consider Single-look (SL) unfiltered data, Multilook (ML) unfiltered data, and CAESAR (CA) multilook filtered data. SqueeSAR filtering, which provides a different diadic approximation of $\widehat{\mathbf{C}}_x$ compared to CAESAR, is not considered hereafter in light of the results of detection comparisons carried out in [33].

The combinations of the considered model and data components lead to the detection schemes summarized in Table I and related, when possible, to the available literature. Specifically, the three FCD schemes correspond to the SL-GLRT, MGLRT, and CAESAR-D detectors, which have been previously derived in [8], [27], and [33], respectively. When considering the PCD schemes, it is important to note that the estimate in (8) is no longer optimal in the Maximum Likelihood sense. Additionally, deriving the GLRT detector in a closed form, as in (7), becomes more challenging for these schemes. For this reason, in [8] the Rao test criterion was adopted to derive

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TABLE I: Detection Schemes Considered in the Analysis. Detectors derived in this work are identified as "New scheme" .

Scheme	Data	Model	Literature
SL-FCD	Single-Look	Fully Coherent	SL-GLRT in [8]
SL-PCD	Single-Look	Partially Coherent	Rao test in [30]
ML-FCD	multilook	Fully Coherent	MGLRT in [27]
ML-PCD	multilook	Partially Coherent	New scheme
CA-FCD	CAESAR-filtered	Fully Coherent	CAESAR-D in [33]
CA-PCD	CAESAR-filtered	Partially Coherent	New scheme

the PCD scheme in the case of single-look unfiltered data, which led to the SL-PCD specification of (7), as shown in Eq. (31) in [30]. The derivation of the unfiltered multilook formulation of the decision rule (7) as Rao test for the ML-PCD is provided in Appendix. Interestingly, the equivalence between the GLRT and the Rao tests demonstrated in the SL case for fully coherent scatterers, see [8], is valid also for the ML versions in light of [27] and the provided Appendix. Conversely, as for the SL case, in the presence of scatterer decorrelation the derived ML-PCD cannot be considered as a GLRT detector. Finally, the CA-PCD scheme has been derived with an empirical approach similar to that carried out in [25] for the FCD scenario, that is, by using the covariance matrix obtained from the filtered multilook data within the PCD formulation of (7). Yet, due to the deviation of the filtered data from the Gaussian distribution, the CAESAR specification of (7) no longer coincides with the Rao test.

III. Structure Matrix Model for Decorrelated Scatterers and Related Parameters Estimation

The analysis carried out in the previous Section does not assume any specific structure for the matrix $\mathbf{M}$, which characterizes the scatterer decorrelation properties. While simulations allow a free set up of $\mathbf{M}$ for data generation, the structure matrix must be unavoidably estimated when the decision rule (7) is applied to real data. In principle, a pixel-dependent matrix $\mathbf{M}$ could be adopted for the DS detection over the whole scene. In this approach, the detector would need to be updated pixel-by-pixel to evaluate the test statistic on the left hand side of (7), hereafter denoted by S. More important, in practical applications, it is essential to maintain a constant Probability of False Alarm (PFA), defined as:

$$P_{\text{fa}} = \Pr(S > T|\mathcal{H}_0) \tag{9}$$

where $\Pr(\cdot)$ denotes the probability of an event. To achieve this, the detection threshold T must be updated on a pixel-by-pixel basis as well. Though, since the threshold is numerically determined trough Monte Carlo techniques, implementing such a strategy typically involves a considerable computational cost. A more manageable approach involves the use of a pixel-independent matrix $\mathbf{M}$, applicable to the entire image. This strategy is for instance employed in the full-resolution detection carried out by the SBAS approach, which sets the entries of $\mathbf{M}$ to either 0 or 1 to exclude or include specific interferometric phase differences (corresponding to the phases of $\boldsymbol{x}\boldsymbol{x}^H$). An alternative, adopted in this work, is to use a parametric model to describe the scatterer decorrelation, providing a different way for handling the variability of $\mathbf{M}$ across the image. In particular we account for an exponential temporal decorrelation and a triangular shaped spatial decorrelation, and suppose that no additional sources of decorrelation affect the scatterer. Accordingly, said $\{\cdot\}_{i,j}$ the operator taking the (i,j) entry of a matrix, we assume:

$$\{\mathbf{M}(\tau, b_{cr})\}_{i,j} = \exp\left(-\frac{|t_i - t_j|}{\tau}\right) \Lambda\left(\frac{|b_i - b_j|}{b_{cr}}\right) \tag{10}$$

where τ denote the coherence time and b_{cr} the critical baseline associated with the pixel under test, both depending on the scatterer properties and the system characteristics. Additionally,

$$\Lambda(x) = \begin{cases} 1 - |x|, & |x| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

is the triangular function. With this model, the PCD schemes reduce to FCD schemes in the limit case of $\tau \to \infty$ and $b_{cr} \to \infty$.

In operational scenarios, temporal and spatial decorrelation may be coupled through the interaction of different scattering mechanisms [34]. Furthermore, more sophisticated models accounting for the periodic decorrelation of scatterers commonly observed in real scenes could also be considered [35]. Nevertheless, our objective is not to provide an "off-the-shelf" detection solution, but rather to demonstrate the importance of interferogram weighting for the detection of decorrelating scatterers. For this reason, we deliberately adopted the simplest and most widely used decorrelation model in the literature, which accounts for the spatial and temporal diversity of the acquisitions.

Next, we provide the guidelines for estimating the decorrelation parameters τ and b_{cr}, highlighting the main critical aspects and intrinsic limitations of the procedure. The proposed parameter estimate is based on the use of the coherence matrix $\mathbf{\Gamma}_x$, which relates to the covariance matrix $\mathbf{C}_x$ as follow:

$$\{\mathbf{\Gamma}_x\}_{i,j} = \frac{\{\mathbf{C}_x\}_{i,j}}{\sqrt{\{\mathbf{C}_x\}_{i,i}}\sqrt{\{\mathbf{C}_x\}_{j,j}}} \tag{11}$$

A measure $\widehat{\mathbf{\Gamma}}_x$ of $\mathbf{\Gamma}_x$ is given by (11) substituting $\mathbf{C}_x$ with the sample covariance matrix:

$$\widehat{\mathbf{C}}_x = \frac{1}{L}\mathbf{X}\mathbf{X}^H$$

where $\mathbf{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L]$ collects L independent and identically distributed data looks.

Based on the expression (6) of $\mathbf{C}_x$, the coherence matrix is modeled as:

$$\mathbf{\Gamma}_x = \tilde{\mathbf{A}}(s,v) \odot \left[\frac{\sigma_0^2}{\sigma_0^2 + \sigma_n^2}\mathbf{M}(\tau, b_{cr}) + \frac{\sigma_n^2}{\sigma_0^2 + \sigma_n^2}\mathbf{I}_N\right] \tag{12}$$

where the matrix $\tilde{\mathbf{A}}(s,v) = N\mathbf{A}(s,v)$ has diagonal entries all equal to 1 and $\sigma_0^2 = \sigma_\gamma^2/N$ is the variance of the elements of the signal component $\boldsymbol{a}(s,v) \odot \boldsymbol{\gamma}$ in (1). Since $\tilde{\mathbf{A}}(s,v)$ consists of complex exponential terms (with unit modulus) and

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$\mathbf{M}(\tau, b_{cr})$ contains only real, non negative entries, it follows that:

$$|\mathbf{\Gamma}_x| = \frac{\sigma_0^2}{\sigma_0^2+\sigma_n^2}\mathbf{M}(\tau, b_{cr}) + \frac{\sigma_n^2}{\sigma_0^2+\sigma_n^2}\mathbf{I}_N \tag{13}$$

where $|\cdot|$ is the element wise absolute value operator. The model in (13) depends on the four unknown parameters τ, b_{cr}, σ_0^2 and σ_n^2, which can be tuned to fit the measure $|\widehat{\mathbf{\Gamma}}_x|$.

In principle, the four unknowns could be estimated by solving a Separable Nonlinear Least Square (SNLS) problem, with (σ_0^2, σ_n^2) as linear parameters and (τ, b_{cr}) as non linear parameters [36], thus minimizing the cost function:

$$J(\sigma_0^2, \sigma_n^2, \tau, b_{cr}) = \| |\mathbf{\Gamma}_x| - |\widehat{\mathbf{\Gamma}}_x| \|_F^2 \tag{14}$$

where $\|\cdot\|_F$ denotes the Frobenius norm operator. However, in regions characterized by significant decorrelation, the coherence matrix may be affected by structural ambiguities. Specifically, when decorrelation is strong, the structure matrix tends to approach the identity matrix. As a result, the off-diagonal elements of the coherence matrix, originated solely by the signal component, become very weak compared to the diagonal elements, which contain contributions from both the signal and noise power. Consequently, the adopted model becomes largely insensitive to the nonlinear decorrelation parameters. In other words, high decorrelation level and low signal-to-noise ratio values produce similar effect on the coherence matrix, thus making it difficult to reliably estimate the unknown parameters.

A way to circumvent this problem relies on the use of additional information to achieve an estimate $\widehat{\sigma_n^2}$ of the noise power, which, differently from the decorrelation parameters, is not pixel-dependent. This estimate can then be exploited to solve the SNLS problem with a reduced number of unknowns. More specifically, the estimates $\widehat{\sigma_0^2}$ and $\widehat{\sigma_n^2}$ can be substituted in (13), with

$$\widehat{\sigma_0^2} = \frac{\mathrm{tr}\left(\widehat{\mathbf{C}}_x\right)}{N} - \widehat{\sigma_n^2} \tag{15}$$

according to the total power constraint

$$\sigma_0^2 + \sigma_n^2 = \frac{\mathrm{tr}\left(\mathbf{C}_x\right)}{N} \tag{16}$$

In this way, neglecting the terms independent from the two residual unknown (non linear) parameters, the cost function (14) can be rewritten as:

$$J(\tau, b_{cr}) = \alpha \|\mathbf{M}(\tau, b_{cr})\|_F^2 - 2\langle \mathbf{M}(\tau, b_{cr}), |\widehat{\mathbf{\Gamma}}| \rangle_F \tag{17}$$

where $\langle \cdot, \cdot \rangle_F$ is the Frobenius inner product operator and

$$\alpha = \frac{\widehat{\sigma_0^2}}{\widehat{\sigma_0^2} + \widehat{\sigma_n^2}} \tag{18}$$

In this work we have simply set $\widehat{\sigma_n^2} = 0$. From (13), it is evident that this assumption implies that the structure matrix and the coherence matrix are identical. Finally, it is important to note that, although this assumption preserves the overall trend of the estimated parameters, it may cause a slight underestimation of their actual values (or equivalently, an overestimation of the corresponding decorrelation effect), particularly for the temporal decorrelation.

IV. Performance Assessment

In this section we analyze and compare the detection and estimation capability of the schemes considered in Table I. The detection analysis is carried out in terms of probability of detection

$$P_{\mathrm{d}} = \Pr(S > T | \mathcal{H}_1) \tag{19}$$

achieved with a fixed probability of false alarm P_{fa}, defined as in (9). Both probabilities are evaluated by resorting to Monte Carlo techniques, over 10^7 (for the P_{fa}) and 5×10^3 (for the P_{d}) realizations of simulated data. Finally, the estimation accuracy of the PoI is also compared in terms of root mean square error (rmse).

The data are generated by exploiting the system parameters of a real COSMO-SkyMed stack of $N = 38$ acquisitions, whose distribution in the temporal/spatial baseline domain is reported in Fig. 1.

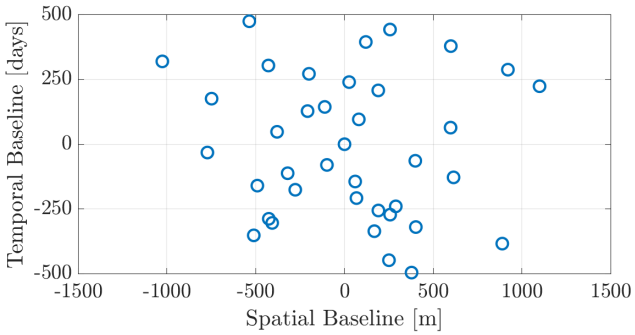


Fig. 1: Baseline distribution.

With the sole aim of analyzing the results with respect to a single coherence parameter, thus avoiding report them separately for the two decorrelation parameters influencing the structure matrix, we introduce the *structure index*, defined as:

$$\rho(\tau, b_{cr}) = \frac{\|\mathbf{M}(\tau, b_{cr})\|_F}{\|\mathbf{M}(+\infty, +\infty)\|_F} \tag{20}$$

This index runs from $1/\sqrt{N}$ to 1, which are the values corresponding to a fully incoherent and a fully coherent scatterer, respectively, and represents a global measure of the overall level of the scatterer coherence. The level ρ_0 of the index (20) corresponds to all pairs (τ, b_{cr}) belonging to the continuous set

$$\mathcal{D}(\rho_0) \equiv \left\{(\tau, b_{cr}) \in \mathbb{R}_0^+ \times \mathbb{R}_0^+ : \rho(\tau, b_{cr}) = \rho_0\right\} \tag{21}$$

where $\mathbb{R}_0^+$ is the set of non negative real numbers.

From an operative point of view, the continuous set (21) can be approximated with the finite set:

$$\mathcal{D}(\rho_0) \equiv \{(\tau, b_{cr}) \in \mathcal{G} : \rho_0 - \epsilon \le \rho(\tau, b_{cr}) \le \rho_0 + \epsilon\} \tag{22}$$

where $\mathcal{G}$ is a uniformly spaced 2D grid covering the coherence time/critical baseline domain, defined according to the temporal and spatial baseline distributions. The value of ϵ is determined empirically, taking into account two requirements: ensuring that the set $\mathcal{D}(\rho_0)$ has a sufficiently large cardinality and maintaining the structure index approximately constant to ρ_0 within this set.

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Under $\mathcal{H}_1$, the simulated data have been generated assuming the presence of a DS in the position $(s, v) = (0, 0)$ of the RT/DMV domain, which decorrelates according to four values of the structure index, $\rho_0 \in \{0.3, 0.5, 0.7, 1\}$. For these value, the set $\mathcal{D}(\rho_0)$ in (22) has been defined with $\epsilon = 5 \times 10^{-3}$ and $\mathcal{G}$ spanning the interval $[10, 2000]$ days in the τ dimension with a spacing of 30 days and the interval $[50, 5000]$ meters in the b_{cr} dimension with a spacing of 100 meter.

The processing has been carried out by applying the detectors listed in Table I to the simulated data. The multilook schemes have been implemented with a number of looks $L = 4$. The estimates (8) of the scatterer parameters have been performed by discretizing the RT/DMV domain with a uniform 2D grid spanning the intervals $[-50, 50]$ meters in the RT dimension and $[-2, 2]$ cm/year in the DMV dimension, with grid spacing of 1 meter and 0.1 cm/year, respectively, set according to the Rayleigh resolutions, [13]. The data simulated under $\mathcal{H}_0$ have been processed for each $(\tau, b_{cr}) \in \mathcal{G}$ and in the limit case $(\tau, b_{cr}) \to (+\infty, +\infty)$ to evaluate the correspondence between P_{fa} and T for the PCD and FCD schemes, respectively. The detection thresholds have been subsequently set to achieve a $P_{\text{fa}} = 10^{-5}$. To illustrate how T varies with the decorrelation parameters, the surface $T(\tau, b_{cr})$ of the ML-PCD scheme evaluated over $\mathcal{G}$ and corresponding to the fixed P_{fa} is shown in Figure 2. The figure indicates that, at least in the regions characterized by a not-negligible level of correlation, the detection threshold required to maintain the specified false alarm rate increases as the correlation parameters grow. The surface exhibits a rather regular pattern, with some irregularities attributable to the grid configuration. In fact, the grid has been selected to balance the contrasting requirements of accuracy and computational burden. The latter is indeed directly related to the number of false alarm probability curves required. Under the assumption of either (filtered or unfiltered) SL processing or unfiltered ML processing carried out with the same number of looks across the entire scene, this number corresponds to the number of bins in the decorrelation grid. Therefore, we can conclude that the number of bins in the decorrelation grid determines the increase in computational burden of the PCD schemes compared to the FCD schemes, which, under the same assumption, require only the evaluation of a single false alarm curve.

For the evaluation of P_{d}, the set $\mathcal{D}(\rho_0)$ used to simulate data under $\mathcal{H}_1$ is assumed to be known. We point out that, since the probability of detection generally varies across the decorrelation parameters providing a given ρ_0, the performance of the detection schemes has been assessed by averaging the P_{d} values over $\mathcal{D}(\rho_0)$.

In Fig. 3, the averaged P_{d} of the schemes in Table I are plotted versus the Signal-to-Noise Ratio (SNR) defined as:

$$\text{SNR} = \frac{\sigma_0^2}{\sigma_n^2} = \frac{\sigma_\gamma^2}{N\sigma_n^2} \tag{23}$$

The Figure illustrates first of all that, under conditions of severe decorrelation as in Fig. 3a, only the ML-PCD scheme, which uses unfiltered multilook data and leverages a properly-designed correlation model, is capable of achieving the maximum detection probability ($P_{\text{d}} = 1$) within the analysed SNR

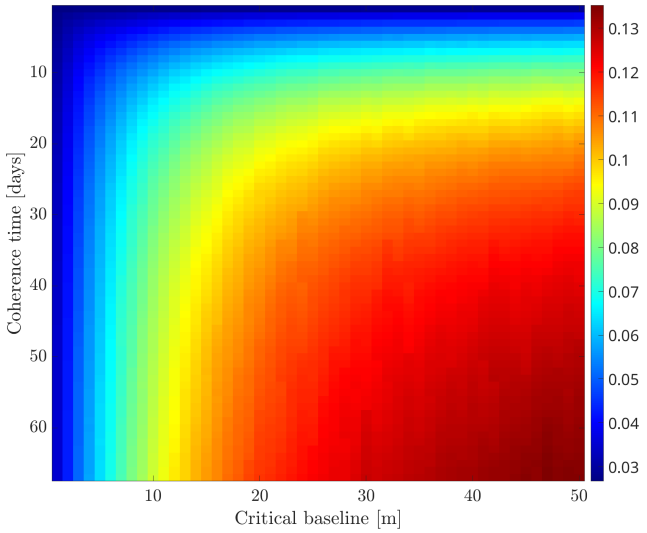


Fig. 2: Detection threshold of ML-PCD over the grid $\mathcal{G}$ for $P_{\text{fa}} = 10^{-5}$.

range. The absence of one or both of these features leads to a performance degradation, although high detection probability ($P_{\text{d}} \geq 0.8$) is still attainable when at least one features is employed, as evidenced by the CA-PCD and ML-FCD curves. Secondly, as expected, the performance of all PCD schemes for DS detection approaches those of the corresponding FCD schemes for increasing ρ_0, see Fig. 3d. It is worth to note that, at least with the assumed system parameters, the improvements of the PCD over the corresponding FCD schemes becomes quantitatively less relevant already at $\rho_0 = 0.5$.

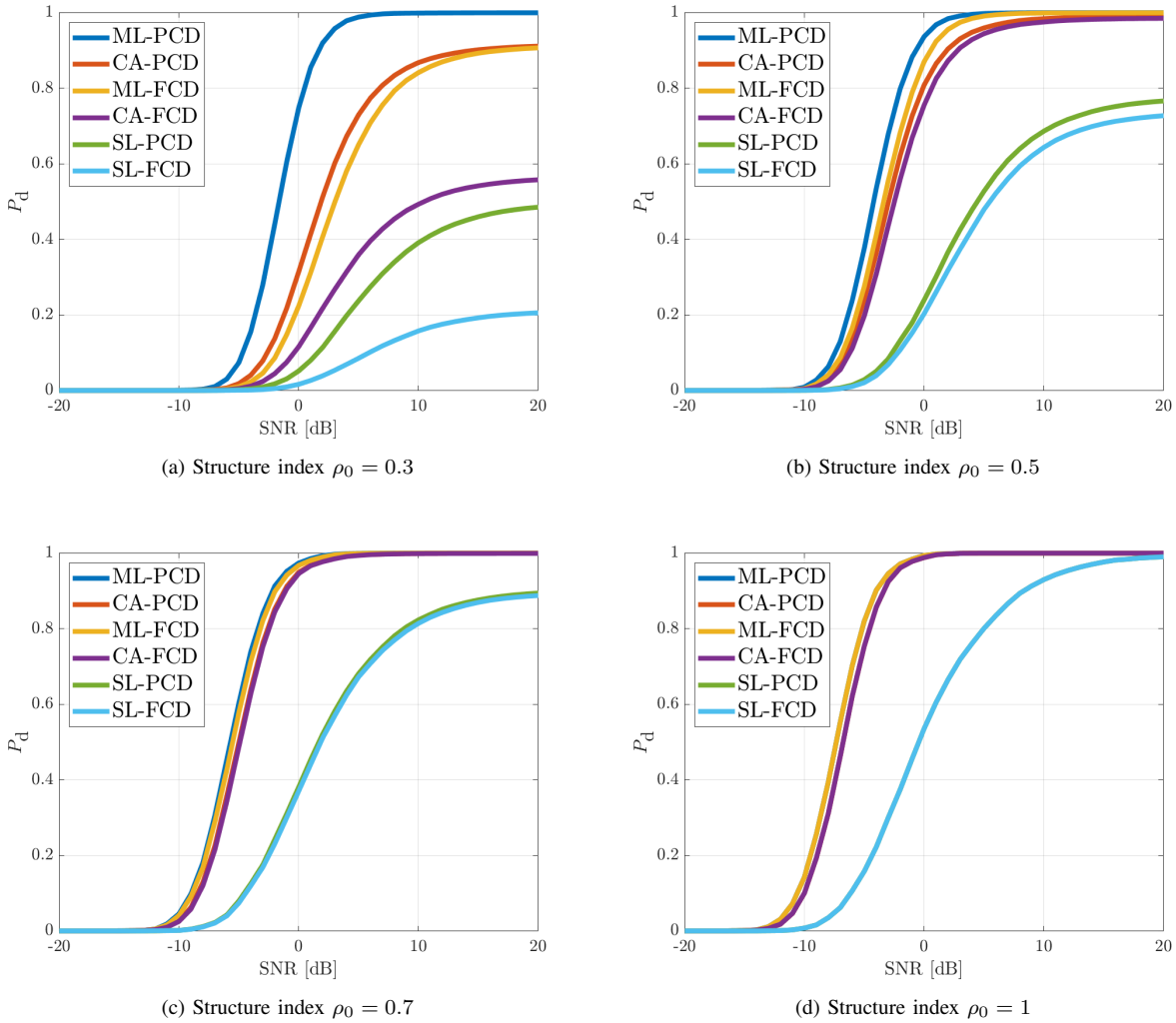
In the transition from Fig. 3a to Fig. 3b it can be seen that an inversion in the detection performances occurs between the CA-PCD and ML-FCD schemes with increasing scatterer coherence. This behavior is likely the result of a tradeoff between the greater accuracy of the decorrelation model used by CA-PCD and the more complete (unfiltered) dataset processed by ML-FCD. Note that this finding is in agreement with the results in [25], which showed that the ML-FCD upper bounds the CAESAR detector for the limiting case of fully coherent model.

Finally, with the exception of ML-PCD, all the remaining detectors exhibit a coherence gain (defined as the reduction in the required SNR to achieve a given P_{d} as the tested DS becomes more coherent) that results in not only a shift of the detection curves toward lower SNR values but also variations of their shape. These curve variations highlight the inability of these detectors to achieve $P_{\text{d}} = 1$ at low ρ_0 values, even under high SNR conditions. This effect is particularly pronounced in the case of single-look schemes, which only approach the maximum detection probability asymptotically with ρ_0. In contrast, over the considered values of ρ_0, the P_{d} curve for ML-PCD is only characterized by a leftward shift driven by the coherence index increase.

The final part of this Section is devoted to the analysis of the PoI estimation accuracy achieved by the considered detection

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(a) Structure index $\rho_0 = 0.3$

(b) Structure index $\rho_0 = 0.5$

(c) Structure index $\rho_0 = 0.7$

(d) Structure index $\rho_0 = 1$

Fig. 3: Probability of detection versus Signal-to-Noise Ratio.

schemes in terms of rmse. In Figures 4a and 4b the rmse of the estimated velocity and topography is plotted versus the SNR for a structure index $\rho_0 = 0.5$. As expected, the curves show a slight improvement in the estimation capability of the partially coherent strategies, tuned for the decorrelation degree of the data, with respect to the corresponding fully coherent counterparts.

V. Experimental Results on Real Data

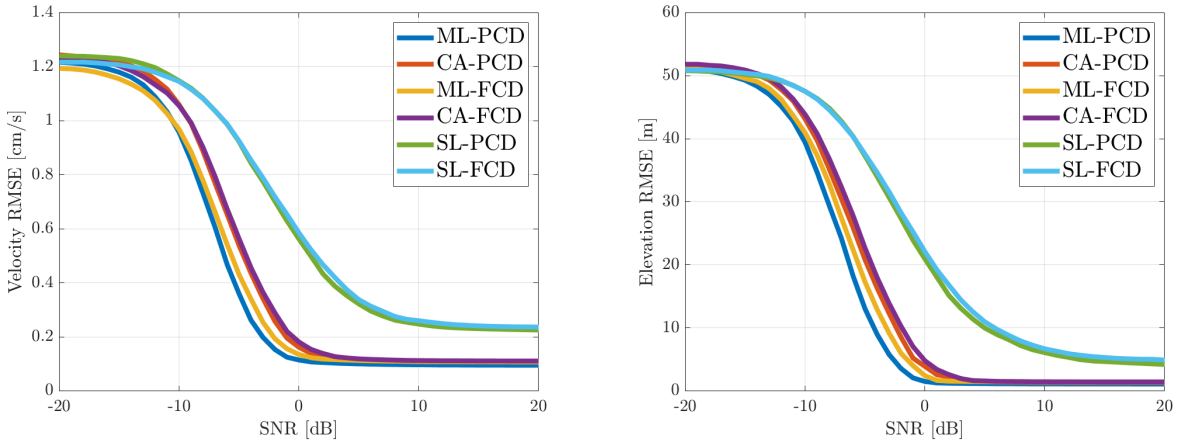
To evaluate the effectiveness of the proposed detection strategies, the schemes listed in Table I have been applied also to the real COSMO-SkyMed dataset introduced in Section IV for the simulated data. Further characteristics and parameters of this dataset are summarized in Table II. The (temporally-averaged) amplitude image is shown in Figure 5. The test region corresponds to the town of Sant'Angelo in Formis, located in the northern part of Naples, South Italy. The choice of COSMO-SkyMed data is related to the fact that, differently from Sentinel-1 where very short correlated scatterers act on the same level as the noise [37], they fit well with the additive white noise model assumption in (1) and allow to simplify the comparison of all the detectors under the same false alarm probability [37]. The site has been selected because it includes a mix of sparsely vegetated and urban areas and has been extensively analyzed also in previous papers [25], [26].

Data were preliminary calibrated for atmospheric phase screen compensation with a low-resolution processing. In addition, the area is characterized by rather regular displacement patterns well captured at the low-resolution processing stage. This leads to a DMV pattern which is almost everywhere characterized by the absence of residual displacements, making the dataset suitable for cross comparison of all the presented approaches. This feature is particularly relevant in the absence of independent (ground truth) deformation data characterized by a sufficiently high measurement density and coverage.

A pre-processing step has been performed to estimate the decorrelation parameters τ and b_{cr} on a pixel-by-pixel basis, through the minimization of the cost function (17). The estimate has been carried out by assuming $\widehat{\sigma_n^2} = 0$, although, as discussed in Sec. III, this may introduce an underestimation of the decorrelation parameters, primarily resulting in lower values of τ. The sample covariance matrix $\widehat{\mathbf{C}}_x$ and the corresponding measured coherence matrix $\widehat{\mathbf{\Gamma}}_x$ have been computed using a fixed number of $L = 9$ looks, obtained

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(a) Velocity rmse versus SNR

(b) Topography rmse versus SNR

Fig. 4: Root mean square errors versus Signal-to-Noise Ratio evaluated for a structure index $\rho_0 = 0.5$.

Data-set Parameters	
number of images	38
acquisition interval	January 2017 - September 2019
slant range resolution	1.44 m
azimuth resolution	2.6 m
elevation resolution	5.49 m
DMV resolution	0.59 cm/year
wavelength	3.1 cm
sensor-scene distance	747.92 km
look angle	34.72°
acquisition mode	stripmap
test scene size (az × rg)	1000 × 1300 pixels
orbit direction	ascending

TABLE II: Parameters of the processed dataset

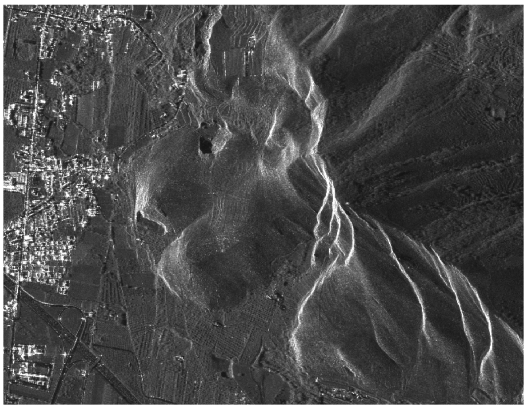


Fig. 5: Temporally averaged amplitude image of the tested region.

with a boxcar window of size 3×3 pixels centered around the processed pixel. Finally, the grid $\mathcal{G}$ defined in Sec. IV has been employed to discretize the coherence time/critical baseline domain. Figure 6a and 6b display the spatial distribution of the estimated τ and b_{cr}, respectively, which are subsequently used in the implementation of PCD schemes. As expected, the highest values of the coherence time are estimated in the urban area. Furthermore, the critical baseline closely follows the topographic variations, which are also reflected in the radiometric modulations evident in the amplitude image shown in Figure 5. Figure 6c presents the distribution of the structure index (20), which synthetically captures the combined effects of both temporal and spatial decorrelation within the imaged scene.

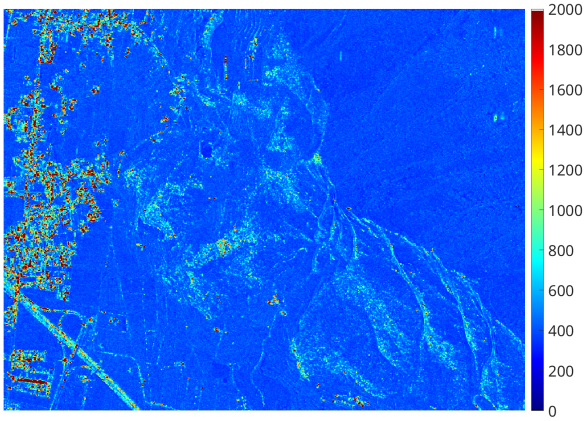
The processing has been performed in the RT/DMV domain, sampled according to the 2D grid defined in Section IV. The detection threshold has been set to ensure a false alarm probability of 10^{-5}. For the multilook schemes, the same boxcar window used during pre-processing ($L = 9$) has been employed. Nevertheless, for these schemes, the detection threshold that guarantees the specified P_{fa} decreases with the number of looks, assuming they are statistically independent [27]. In practice, the looks processed for each pixel are typically partially correlated. To prevent the false alarm probability from exceeding the desired level, the thresholds for the multilook schemes are set based on the equivalent number of looks, L_{eq}, which reflects the actual (usually lower) number of independent looks corresponding to L [38]. It is important to note that L_{eq} is estimated within homogeneous regions of the images, where the radar cross-section is assumed to be constant [38]. In these regions, the equivalent number of looks can vary within a limited range. The minimum estimated value, $L_{\text{eq}} = 4$, has been used to guarantee a P_{fa} of 10^{-5} at most.

Figure 7 illustrates the distribution of the detection threshold used by the ML-PCD scheme, which ensures the specified false alarm probability constraint. The figure highlights that, in accordance with the results obtained from simulated data and described in Sect. IV, regions with higher levels of coherence require higher threshold values.

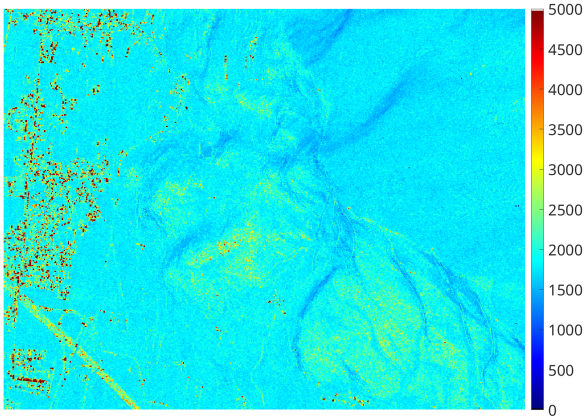
Finally, Figure 8 displays the distribution of detected scatterers achieved by the schemes listed in Table I. The values associated with the detected pixels represent the estimated DMV, measured in cm/year. Based on the colormap, the predominantly green hues across all maps indicate a stable distribution of residual velocity centered around zero. The figure primarily highlights the superior performance of the PCD schemes on the left compared to the FCD schemes on the right in terms of the number of detections. Furthermore,

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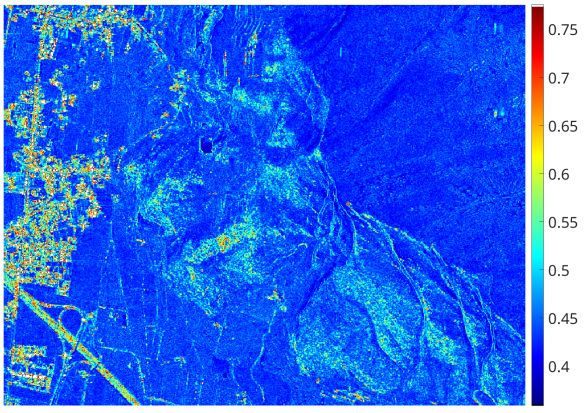
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(a) : Coherence time - τ [days]



(b) : Critical Baseline - b_{cr} [m]



(c) : Structure Index - $\rho(\tau, b_{cr})$

Fig. 6: Spatial distribution of the (top) estimated coherence time, (middle) critical baseline, (bottom) structure index.

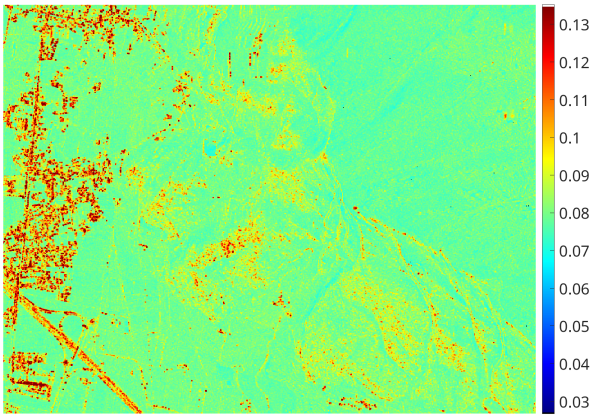


Fig. 7: Detection threshold of ML-PCD over the grid $\mathcal{G}$ for $P_{\text{fa}} = 10^{-5}$.

a more detailed discussion of these results is provided in the subsequent section.

VI. Result Discussions and Open Issues

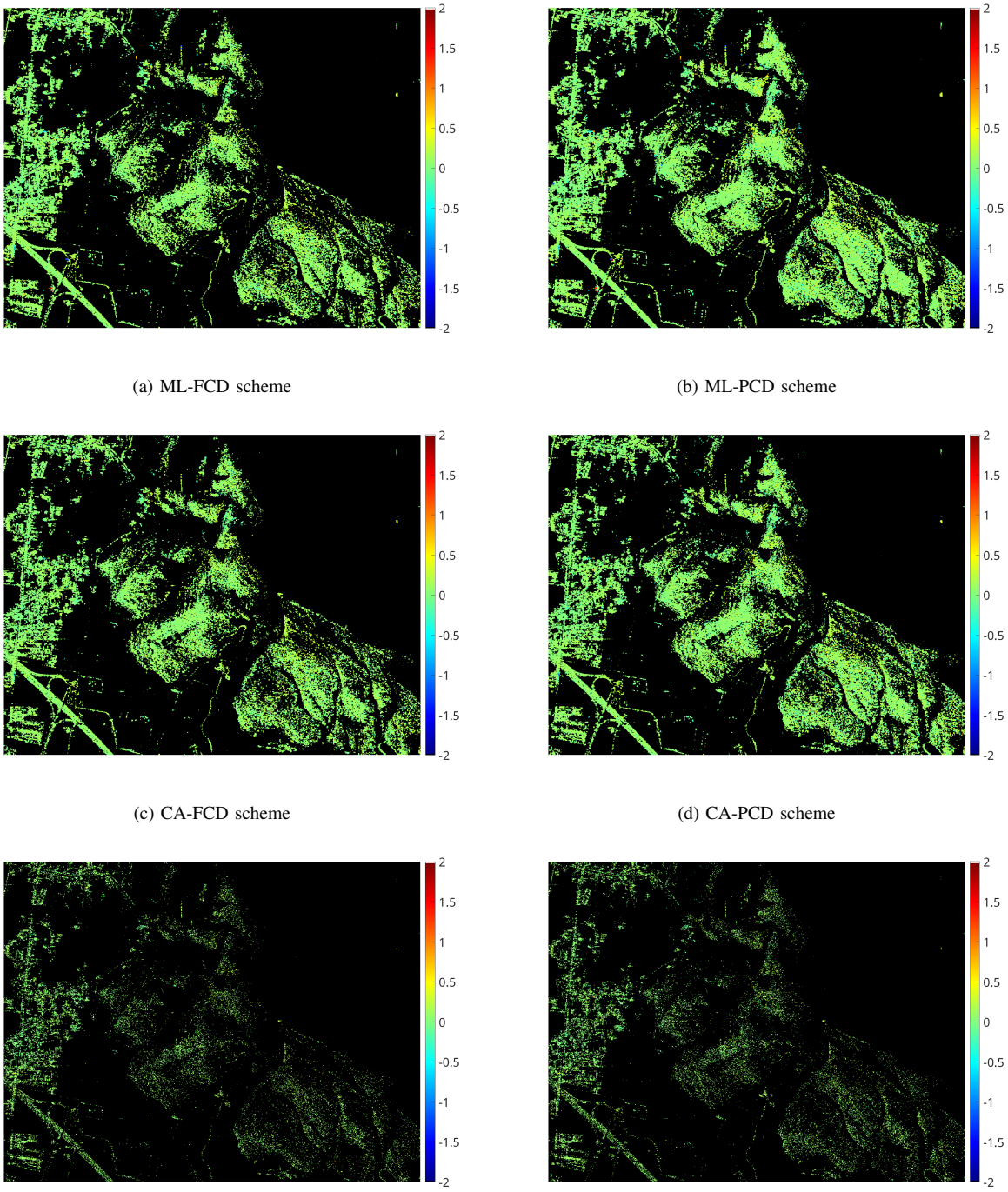
As mentioned earlier, Figure 8 presents the distribution of the scatterers detected using the different PCD and FCD strategies. A key observation, consistent with the results obtained from simulated data discussed in Sec. IV and supported by the literature [25], concerns the importance of employing a multilook approach, either with filtering (as in the CAESAR-based detector) or without filtering, to significantly increase the density of detected DSs. This is demonstrated by comparing Figures 8b-8c with Figures 8f and 8e. Furthermore, incorporating the scene's coherence properties into the detector design leads to additional detection performance improvements, as shown by the comparison between the figures in the left and right columns. Among all the analyzed approaches, the ML-PCD achieves the best performance. Its ability to account for the decorrelation model, combined with the processing of unfiltered multilook data, enables superior results, see Figure 8b.

To highlight the differences among the results of the considered detection schemes, Figure 9 shows the pixels exclusively detected by the FCD schemes (left column) and the PCD schemes (right column). The colormap represents the structure index (20). For improved visualization, the detected pixels have been expanded with the four surrounding pixels. The increased number of scatterers detected by the PCD schemes, particularly in regions characterized by low correlation, is clearly evident.

A numerical summary of the detection results is also provided in tabular form. Table III reports the number of scatterers detected by each of the six analyzed schemes. Table IV shows the reciprocal increase/decrease detection percentages for each scheme relative to all others, enabling a direct comparison of their relative performance in terms of detection gains. Both tables rank the schemes based on the number of detections and quantify the performance improvement of each method compared to the others. Notably, ML-PCD achieves a 21.3%

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(a) ML-FCD scheme

(b) ML-PCD scheme

(c) CA-FCD scheme

(d) CA-PCD scheme

(e) SL-FCD scheme

(f) SL-PCD scheme

Fig. 8: Distribution of the scatterers detected using the schemes in Table I. The colomap is set according to the estimated DMV, measured in cm/year.

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increase in detected scatterers compared to the current state-of-the-art method, represented by ML-FCD.

Based on the above Figures and Tables, some final considerations can be provided. Figures 9 and 6 shows that the greatest increase in detections achieved by PCD schemes compared to FCD occurs mainly in regions affected by temporal decorrelation. Notably, even the SL-PCD scheme, which does not benefit from data multilooking, still demonstrates detection improvements in these areas. This suggests that an appropriate weighting of the interferograms, achieved through a suitable design of the detector, may mitigate the effects of decorrelation in DS detection. The need to maintain a constant false alarm rate across the entire scene can lead to a substantial computational burden in setting the detection threshold for schemes using unfiltered multilook data and/or designed according to the PCD model.

As for the unfiltered multilook approach, this issue was previously addressed in [27] where the dependence of the ML-FCD threshold providing a specified false alarm probability on the number of looks was demonstrated. This leads to the necessity of using an adaptive threshold in multi-resolution scenarios, where the number of looks vary on a pixel-by-pixel basis to preserve the spatial details of the observed scene. Under these circumstances, although the ML-PCD scheme detects the highest number of scatterers, the CA-PCD scheme may be preferable. In fact, following the approach outlined in [25], it can be demonstrated that, similar to the CA-FCD scheme, CA-PCD can ensure constant false alarm rate with a detection threshold independent of the number of employed looks, and coinciding with that of the SL-PCD detector. Moreover, as shown in Table IV, although it detects fewer scatterers compared to the ML-PCD detector, it still outperforms the current state-of-the-art approaches.

Concerning the PCD design, the dependence of the detection threshold on the parameters of the adopted correlation model (such as coherence time and critical baseline) has been demonstrated through both simulated and real data, as shown in Figures 2 and 7. Therefore, PCD schemes require a variable detection threshold to maintain a constant false alarm rate even when a fixed number of looks or a filtered multilook approach are used, thus resulting in increased computational costs. Possible strategies to mitigate this burden include modeling the threshold surface corresponding to the fixed level of false alarm over the decorrelation parameter domain to analytically set the threshold without relying on Monte Carlo simulations. Alternatively, as done (with an "on-off" interferogram weighting strategy) in [9], a trade-off between computational complexity and detection/estimation accuracy could be achieved by adopting a global, pixel-independent setting of the parameters describing the decorrelation, thus enabling the use of a single detection threshold.

The results achieved by the PCD schemes in terms of both the number of detections and the accuracy of the PoI estimates are strictly related to the exploited scattering correlation model and, on real data, to the estimation of its parameters. The model presented in (10) was chosen for its simplicity, with the primary aim of demonstrating the importance of weighting interferograms in the detection of DS. Of course, alternative models can be utilized, allowing a trade-off between model accuracy, detection performance, and computational complexity. As for the estimation of the model parameters (specifically, the coherence time and the critical baseline), we have previously emphasized that assuming $\widehat{\sigma_n^2} = 0$ to circumvent the structural ambiguities of the coherence matrix in (13) may lead to an underestimation of the decorrelation parameters. In fact, additional experiments, whose results are not presented here for brevity, have shown that increasing the value of $\widehat{\sigma_n^2}$ results in larger values of the estimated decorrelation parameters, particularly the coherence time. This increase, in turn, causes a reduction in the number of detections achieved by the PCD schemes. For the other extreme case corresponding to a noise power equal to the minimum data power across all pixels, the decrease in the number of detections remains very limited, less than 2%.

Regarding the accuracy of the PoI estimates, it is important to note that in cases of DS affected by a high degree of decorrelation, the estimation errors are more likely to occur, which can compromise the physical relevance of the retrieved parameters. This underscores the need to balance detection performance with the reliability of parameter estimates. Addressing these aspects is, however, beyond the scope of the current work. Future research will aim to investigate the dependence of the detection threshold on decorrelation parameters and to analyze the impact of various decorrelation models on the estimation of scatterer PoI. The goal is to not only enhance the detection rate but also to ensure the physical consistency and reliability of the estimated parameters.

	ML	**CA**	**SL**
PCD	308014	273763	72147
FCD	253854	243657	65282

TABLE III: Number of detected scatterers.

	ML-PCD	**CA-PCD**	**ML-FCD**	**CA-FCD**	**SL-PCD**	**SL-FCD**
ML-PCD	0%	12.5%	21.3%	26.4%	327%	372%
CA-PCD	-12.5%	0%	7.8%	12.3%	279%	319%
ML-FCD	-21.3%	-7.8%	0%	4.1%	252%	289%
CA-FCD	-26.4%	-12.3%	-4.1%	0%	238%	273%
SL-PCD	-327%	-279%	-252%	-238%	0%	10.5%
SL-FCD	-372%	-319%	-289%	-273%	-10.5%	0%

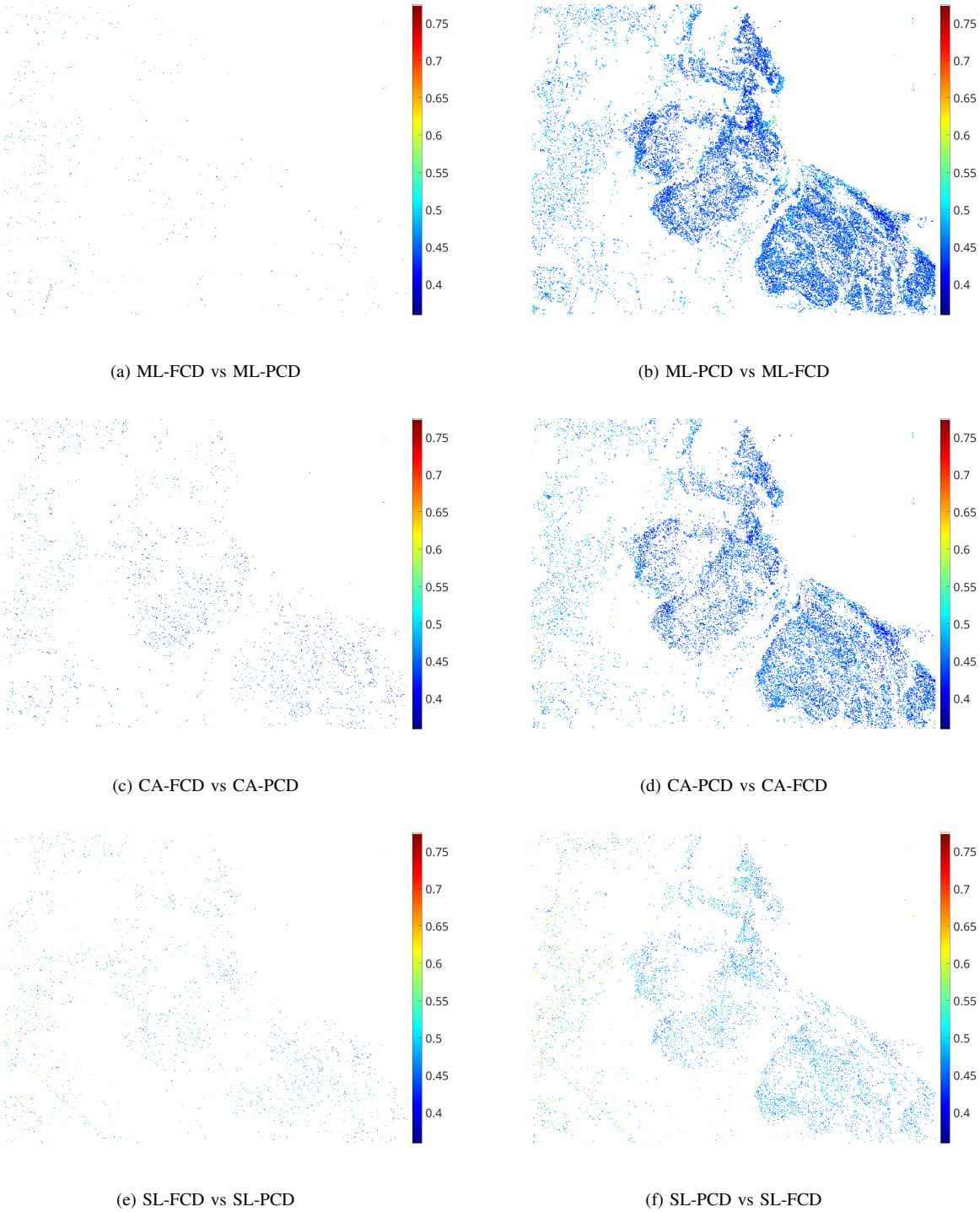
TABLE IV: Overview of the relative performances of the analyzed detectors: each value from a (line,column) position indicates the relative performance of the detector from the corresponding line over the detector from the corresponding column.

VII. Conclusion

This work investigated the design and performance of detection schemes for partially coherent scatterers, emphasizing the advantages of jointly leveraging appropriate decorrelation models and multilook processing. In particular, the proposed partially coherent detectors (PCD), implemented with single-look, multilook, and CAESAR-filtered data, explicitly incorporate the correlation properties of scatterers within the detection framework. The analysis of the results obtained by processing a real dataset acquired by the COSMO-SkyMed sensor and the performance assessment achieved from data simulated with

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(a) ML-FCD vs ML-PCD

(b) ML-PCD vs ML-FCD

(c) CA-FCD vs CA-PCD

(d) CA-PCD vs CA-FCD

(e) SL-FCD vs SL-PCD

(f) SL-PCD vs SL-FCD

Fig. 9: Distribution of the scatterers detected exclusively by (left) FCD schemes and (right) PCD schemes. The colormap is set according to the structure index (20).

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the same system parameters, indicate that accounting for the decorrelation at the detection stage can lead to an increase in the number of detected scatterers for a fixed false alarm rate. The extension to the analysis of SAR sensors characterized by different samplings of the spatial-temporal baseline domain is beyond the scope of the present analysis and is reserved for future developments.

VIII. Acknowledgments

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Appendix

Consider the $N \times L$ data matrix $\mathbf{X}_L = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L]$ collecting the observations (looks) of L independent and identically distributed random vectors. Assume each random vector to be multivariate zero mean complex Gaussian distributed, with covariance matrix under the hypothesis $\mathcal{H}_i$ given by

$$\mathbf{C}_i = \begin{cases} \sigma_n^2 \mathbf{I}_N & i = 0 \\ \sigma_\gamma^2 \mathbf{A}(s,v) \odot \mathbf{M}(\tau, b_{cr}) + \sigma_n^2 \mathbf{I}_N & i = 1 \end{cases} \tag{24}$$

Suppose that the PoI (s, v) and the decorrelation parameters (τ, b_{cr}) are known and assume that the variances σ_γ^2 and σ_n^2 are unknown. Finally, define the vector of the unknown parameters $\boldsymbol{\theta} = [\theta_0, \theta_1]^T$, with $\theta_0 = 0$ under $\mathcal{H}_0$ and $\theta_0 = \sigma_\gamma^2$ under $\mathcal{H}_1$, and $\theta_1 = \sigma_n^2$.

The multilook Rao test for discriminating between the hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$ is the following decision rule [39]:

$$\left\{ \left\{ \mathbf{J}_L(\boldsymbol{\theta})^{-1} \right\}_{1,1} \left[\frac{\partial \ln p_L(\mathbf{X}_L; \boldsymbol{\theta}, \mathcal{H}_1)}{\partial \theta_0} \right]^2 \right\} \Bigg|_{\boldsymbol{\theta} = \widehat{\boldsymbol{\theta}}_0} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{25}$$

where T is the detection threshold, $p_L(\mathbf{X}_L; \boldsymbol{\theta}, \mathcal{H}_i)$ is the likelihood function of the multilook data under $\mathcal{H}_i$, for $i = 0, 1$, $\widehat{\boldsymbol{\theta}}_0 = [0, \widehat{\sigma_n^2}(\mathcal{H}_0)]^T$ is the multilook MLE of $\boldsymbol{\theta}$ under $\mathcal{H}_0$, and $\mathbf{J}_L(\boldsymbol{\theta})$ is the 2×2 Fisher information matrix derived from $p_L(\mathbf{X}_L; \boldsymbol{\theta}, \mathcal{H}_i)$ as:

$$\{\mathbf{J}_L(\theta)\}_{i,j} = -E\left[\frac{\partial^2 \ln p_L(\mathbf{X}_L; \boldsymbol{\theta}, \mathcal{H}_1)}{\partial \theta_i \partial \theta_j} \right] \tag{26}$$

To evaluate (25), the likelihood function under both hypotheses is required. According to the above assumptions, its expression under $\mathcal{H}_i$ is:

$$p_L\left(\mathbf{X}_L; \boldsymbol{\theta}, \mathcal{H}_i\right) = \frac{\exp\left[-\operatorname{tr}\left(\mathbf{C}_i^{-1} \mathbf{X}_L \mathbf{X}_L^H \right) \right]}{\left[\pi^N \det(\mathbf{C}_i) \right]^L} \tag{27}$$

where $\det(\cdot)$ denotes the determinant of a square matrix. Equation (27) can be conveniently rewritten as

$$p_L\left(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_i\right) = \frac{\exp\left[-L \operatorname{tr}\left(\mathbf{C}_i^{-1} \widehat{\mathbf{C}}_L \right) \right]}{\left[\pi^N \det(\mathbf{C}_i) \right]^L} \tag{28}$$

by noting that the dependence of (27) on the data matrix occurs only thorough the sample covariance matrix $\widehat{\mathbf{C}}_L = \mathbf{X}_L \mathbf{X}_L^H / L$. Using (28) under $\mathcal{H}_0$, the multilook MLE $\widehat{\boldsymbol{\theta}}_0 = [0, \widehat{\sigma_n^2}(\mathcal{H}_0)]^T$ is given by:

$$\widehat{\sigma_n^2}(\mathcal{H}_0) = \arg\max_{\theta 1 \geq 0} \left[p_L(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_0) \right] = \frac{\operatorname{tr}\left(\widehat{\mathbf{C}}_L \right)}{N} \tag{29}$$

Furthermore, it can be easily shown that:

$$\ln p_L(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_1) = L \ln p_1(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_1) \tag{30}$$

and

$$\mathbf{J}_L(\boldsymbol{\theta}) = L \mathbf{J}_1(\boldsymbol{\theta}) \tag{31}$$

In (30), $p_1(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_1)$ is the likelihood function of the single-look data under $\mathcal{H}_1$, evaluated in the multilook sample covariance matrix. In (31), $\mathbf{J}_1(\boldsymbol{\theta})$ is the Fisher information matrix derived from $p_1(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_1)$, which results to be equal to that achieved from $p_1(\widehat{\mathbf{C}}_1; \boldsymbol{\theta}, \mathcal{H}_1)$ because of the statistical averaging operation. Accordingly, by substituting (30) and (31) in (25), the multilook Rao test can be rewritten as:

$$\left\{ \left\{ \mathbf{J}_1(\boldsymbol{\theta})^{-1} \right\}_{1,1} \left[\frac{\partial \ln p_1(\widehat{\mathbf{C}}_L; \boldsymbol{\theta}, \mathcal{H}_1)}{\partial \theta_0} \right]^2 \right\} \Bigg|_{\boldsymbol{\theta} = \widehat{\boldsymbol{\theta}}_0} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{32}$$

with $\widehat{\boldsymbol{\theta}}_0$ given by (29).

In [30], the Rao test was derived for the single-look case, that is from (29) and (32) evaluated with single-look sample covariance matrix $\widehat{\mathbf{C}}_1$ instead of $\widehat{\mathbf{C}}_L$. This derivation led to the following decision rule:

$$\left[S(\widehat{\mathbf{C}}_1) - \frac{1}{N} \right]^2 \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{33}$$

where

$$S(\widehat{\mathbf{C}}_1) = \frac{\operatorname{tr}\left[(\mathbf{A} \odot \mathbf{M}) \, \widehat{\mathbf{C}}_1 \right]}{\operatorname{tr}\left(\widehat{\mathbf{C}}_1 \right)} \tag{34}$$

and the dependence of $\mathbf{A}$ on the PoI and of $\mathbf{M}$ on the decorrelation parameters have been omitted for sake of simplicity. Straightforward manipulations lead to the following equivalent test

$$S(\widehat{\mathbf{C}}_1) \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{35}$$

where T is the new detection threshold, belonging to the interval $[0, 1]$. Since the functional form of this result does not depend on the exploited sample covariance matrix, it is still valid also in the multilook case.

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